

Advanced Communication Networks

شبکه‌های ارتباطی پیشرفته

Module 22

Quality of Service (QoS)

کیفیت خدمات (QoS)



Lecture Notes

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1 Why QoS?

The Problem: Not All Traffic Is Equal

Different applications have fundamentally different network requirements:

Service	Delay Tolerance	Loss Tolerance	Bandwidth	Pattern
Voice (VoLTE)	Very Low (<150ms)	Low (<1%)	Low (64 kbps)	Constant
Video Call	Low (<300ms)	Low (<1%)	Medium (1-5 Mbps)	Variable
Video Streaming	Medium (<5s buffer)	Very Low	High (5-25 Mbps)	Bursty
IoT Sensor	High (seconds)	Very Low	Very Low (kbps)	Periodic
Web Browsing	Medium (<1s)	None (TCP retx)	Variable	Bursty
Email	High (seconds)	None (TCP retx)	Low	Bursty
Autonomous Driving	Ultra-Low (<10ms)	Ultra-Low	Medium	Constant

Without QoS:

- Voice calls get choppy during congestion
- Video freezes while background downloads continue
- Emergency services compete with social media
- Industrial control packets delayed by bulk transfers

QoS Goal:

Differentiate traffic and provide appropriate treatment to each service based on its requirements.

2 QoS Evolution Across Generations

2G	3G	LTE (4G)	5G NR
No QoS (all circuit-switched)	4 traffic classes	Bearer-based QoS (QCI + bearer)	Flow-based QoS (5QI + QoS flow)
Same quality for all	Conversational, streaming, interactive, background	9 standard QCIs; GBR / non-GBR; default + dedicated bearers	90 standard 5QIs; GBR / non-GBR / DC-GBR; reflective QoS; fine granularity

2G (GSM): No Real QoS

- All services are **circuit-switched**
- Each call gets a dedicated timeslot → same quality for everyone
- No differentiation needed (only voice + SMS)

3G (UMTS): Four Traffic Classes

Class	Delay	Example
Conversational	Strict (< 150ms)	Voice, video call
Streaming	Moderate (< 1s)	Video streaming
Interactive	Variable	Web browsing
Background	Best-effort	Email, downloads

LTE (4G): Bearer-Based QoS

- Introduced **EPS bearers** with specific QoS parameters
- **QCI** (QoS Class Identifier): standardized classes
- **ARP** (Allocation and Retention Priority)
- **GBR** (Guaranteed Bit Rate) vs **Non-GBR** bearers

5G NR: Flow-Based QoS

- **QoS Flows** identified by **QFI** (QoS Flow Identifier)
- **5QI** replaces QCI with more values
- **Reflective QoS** for simplified UL treatment
- Finer granularity: multiple flows per PDU session

3 LTE Bearer-Based QoS

■ Bearer Concept

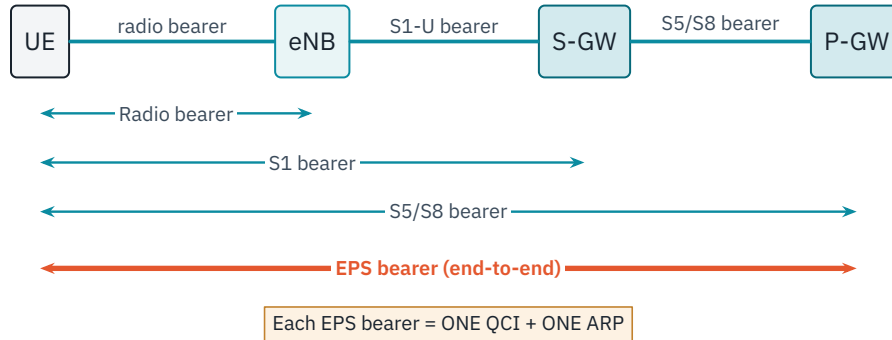


Figure 22.1 The EPS bearer is a chain of radio, S1 and S5/S8 bearers

Bearer = GTP-U tunnel with specific QoS parameters (QCI + ARP + bit rates)

■ Default vs Dedicated Bearers

Aspect	Default Bearer	Dedicated Bearer
Creation	At PDU connectivity (attach)	On-demand (triggered)
Lifetime	Always on (until detach)	Temporary (can be released)
QoS	Best-effort (QCI 9)	Specific QoS (any QCI)
Type	Always Non-GBR	GBR or Non-GBR
IP Address	Assigned with bearer	Shares IP of default
Purpose	Internet access	VoLTE, video, gaming

■ QCI Table (Standardized Values)

QCI	Resource Type	Priority	Packet Delay Budget	Packet Error Loss Rate	Example Services
1	GBR	2	100 ms	10^{-2}	Conversational Voice
2	GBR	4	150 ms	10^{-3}	Conversational Video
3	GBR	3	50 ms	10^{-3}	Real-Time Gaming
4	GBR	5	300 ms	10^{-6}	Non-Conv. Video (Buffered)
5	Non-GBR	1	100 ms	10^{-6}	IMS Signaling
6	Non-GBR	6	300 ms	10^{-6}	Video (Buffered), TCP web
7	Non-GBR	7	100 ms	10^{-3}	Voice, Live Video, Gaming
8	Non-GBR	8	300 ms	10^{-6}	Video (Buffered), TCP web
9	Non-GBR	9	300 ms	10^{-6}	Video, TCP, Email, Chat

■ Mission-Critical (MCX) QCI Values (65-82)

QCI	Resource Type	Priority	Delay	Example
65	GBR	0.7	75 ms	MC-PTT Voice
66	GBR	2	100 ms	MC-Video (non-real-time)
67	GBR	1.5	100 ms	MC-PTT Video
69	Non-GBR	0.5	60 ms	MC-PTT Signaling
70	Non-GBR	5.5	200 ms	MC Data
75	GBR	2.5	50 ms	V2X Messages
79	Non-GBR	6.5	50 ms	V2X Messages
82	GBR	1.9	10 ms	Discrete Automation
83	GBR	2.2	10 ms	Discrete Automation

■ ARP (Allocation and Retention Priority)

ARP determines whether a bearer can be established during congestion:

Parameter	Values	Purpose
Priority Level	1-15 (1=highest)	Relative importance
Pre-emption Capability	Yes/No	Can this bearer preempt others?
Pre-emption Vulnerability	Yes/No	Can this bearer be preempted?

Example Scenario:

Congestion situation - resources are full:

Bearer A: Priority=5, Pre-emption Capability=Yes

Bearer B: Priority=10, Pre-emption Vulnerability=Yes

→ Bearer A can preempt (remove) Bearer B to get resources

→ Emergency calls (Priority=1) can preempt regular calls

■ GBR vs Non-GBR Bearers

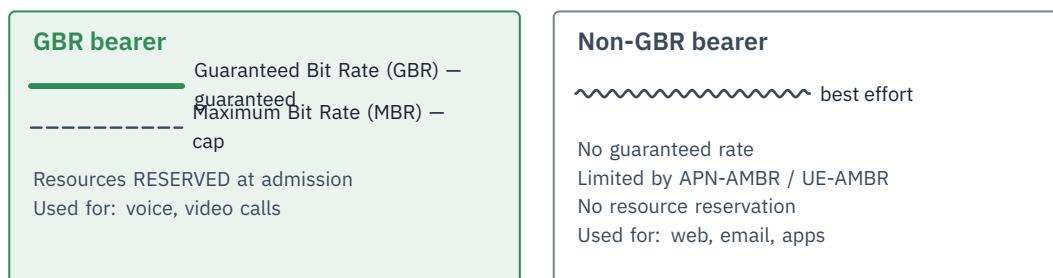


Figure 22.2 GBR and non-GBR bearers

■ LTE QoS Limitations

1. **Coarse granularity:** One QCI per bearer → one bearer per service type
2. **Bearer overhead:** Each new QCI requires a new GTP tunnel
3. **Signaling heavy:** Dedicated bearer setup requires full NAS/S1AP signaling
4. **Inflexible:** Cannot mix traffic with different QoS in one bearer
5. **Limited QCIs:** Only 9 standard values (extended to ~20 with MCX)

4 5G Flow-Based QoS

■ Key Concept: QoS Flow

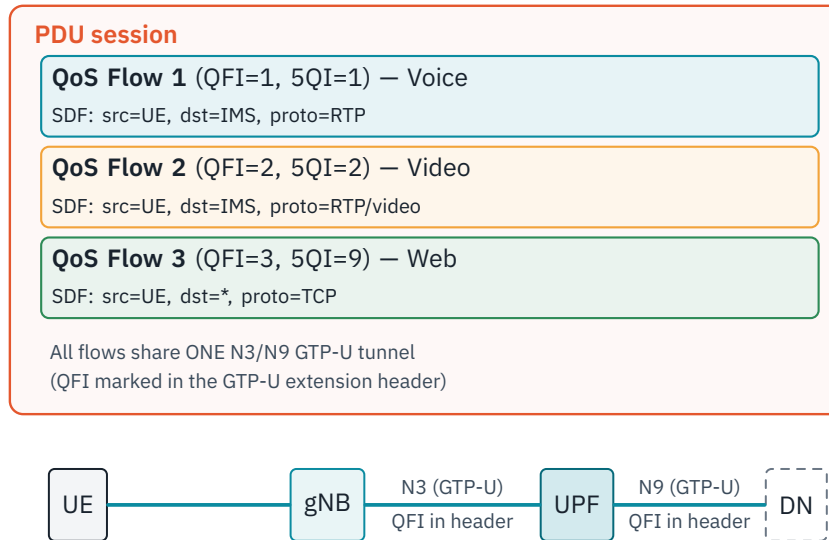


Figure 22.3 5G QoS flows inside one PDU session share one GTP-U tunnel

Key differences from LTE:

- Multiple QoS Flows share ONE PDU session (one GTP tunnel)
- QFI (6-bit) identifies the flow within the tunnel
- No need to create/delete bearers → less signaling
- Finer granularity with less overhead

5QI Table (Key Standardized Values)

5QI	Resource Type	Priority	Packet Delay	Packet Error Rate	Example Services
1	GBR	20	100 ms	10^{-2}	Conversational Voice
2	GBR	40	150 ms	10^{-3}	Conversational Video
3	GBR	30	50 ms	10^{-3}	Real-Time Gaming
4	GBR	50	300 ms	10^{-6}	Non-Conv. Video (Buffered)
5	Non-GBR	10	100 ms	10^{-6}	IMS Signaling
6	Non-GBR	60	300 ms	10^{-6}	Video, TCP (default)
7	Non-GBR	70	100 ms	10^{-3}	Voice, Video, Gaming
8	Non-GBR	80	300 ms	10^{-6}	Video, TCP web
9	Non-GBR	90	300 ms	10^{-6}	Video, TCP, Email (default)
65	GBR	7	75 ms	10^{-2}	MC-PTT Voice
66	GBR	20	100 ms	10^{-2}	Non-MC-PTT Voice
67	GBR	15	100 ms	10^{-3}	MC Video
69	Non-GBR	5	60 ms	10^{-6}	MC-PTT Signaling
70	Non-GBR	55	200 ms	10^{-6}	MC Data
79	Non-GBR	65	50 ms	10^{-2}	V2X Messages
80	Non-GBR	68	10 ms	10^{-6}	Low-Latency eMBB
82	DC-GBR	19	10 ms	10^{-4}	Discrete Automation (small)
83	DC-GBR	22	10 ms	10^{-4}	Discrete Automation (large)
84	DC-GBR	24	30 ms	10^{-5}	Intelligent Transport
85	DC-GBR	21	5 ms	10^{-5}	Electricity Distribution
86	GBR	18	5 ms	10^{-4}	V2X (platooning)
87	DC-GBR	25	5 ms	10^{-3}	Interactive Service (motion)
88	DC-GBR	23	10 ms	10^{-4}	Interactive Service (steering)
89	DC-GBR	26	15 ms	10^{-4}	Visual Content (cloud gaming)
90	DC-GBR	27	20 ms	10^{-4}	Visual Content (AR/ cloud XR)

DC-GBR = Delay-Critical GBR (new in 5G)

QoS Rules

QoS rules tell the UE how to classify uplink traffic:

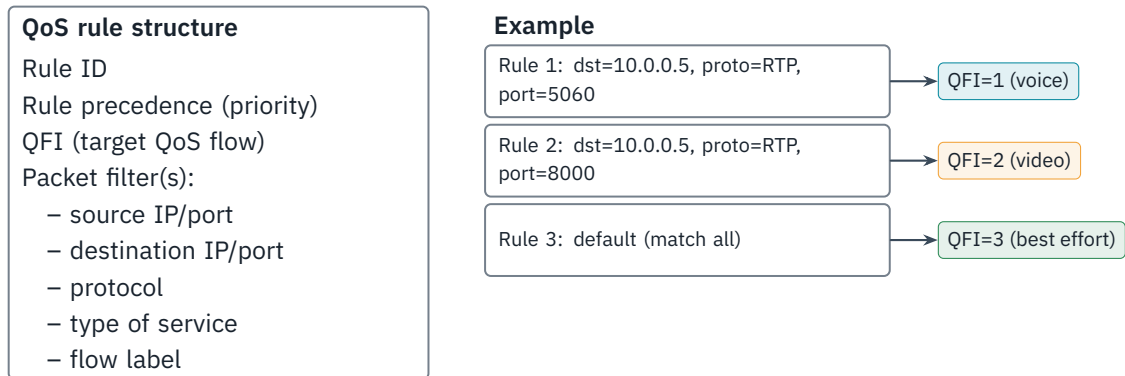


Figure 22.4 Structure of a QoS rule, with example rules

Reflective QoS

1. The network sends a DL packet with QFI = X and RQI = 1 (Reflective QoS Indication).
2. The UE observes the DL packet headers:
 - source IP/port → becomes the UL destination
 - destination IP/port → becomes the UL source
 - QFI = X → the UE uses the same QFI for its reply
3. The UE creates a “derived QoS rule” automatically:
 - no explicit QoS-rule signalling is needed
 - the rule is timer-based: valid for the RQ timer duration

Benefit: less signalling to establish QoS rules. **Use case:** the UE does not know the server IP in advance.

5 QoS Mapping and Enforcement

End-to-End QoS Mapping Diagram

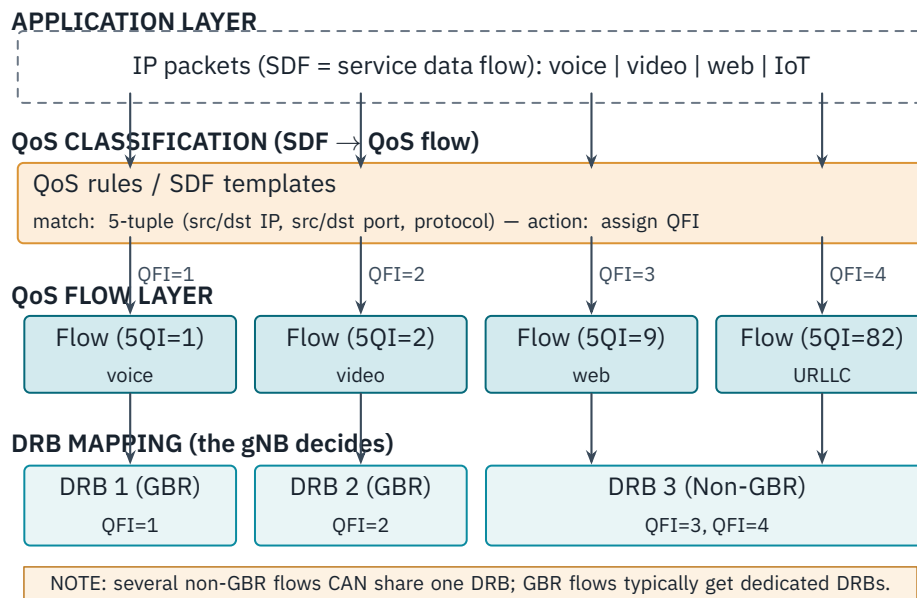


Figure 22.5 End-to-end 5G QoS mapping: SDF → QoS flow → DRB

Role of Each Network Element

Element	QoS Role	Actions
PCF	Policy Decision	Creates PCC rules, determines 5QI/ARP, sends to SMF
SMF	QoS Control	Derives QoS profiles, installs QoS rules, configures UPF
UPF	DL Enforcement	SDF detection, QFI marking, gating, policing, rate limiting
gNB	Radio Scheduling	Maps QoS flows→DRBs, schedules based on 5QI, enforces AMBR
UE	UL Classification	Applies QoS rules, maps UL packets to QoS flows

QoS Enforcement Architecture

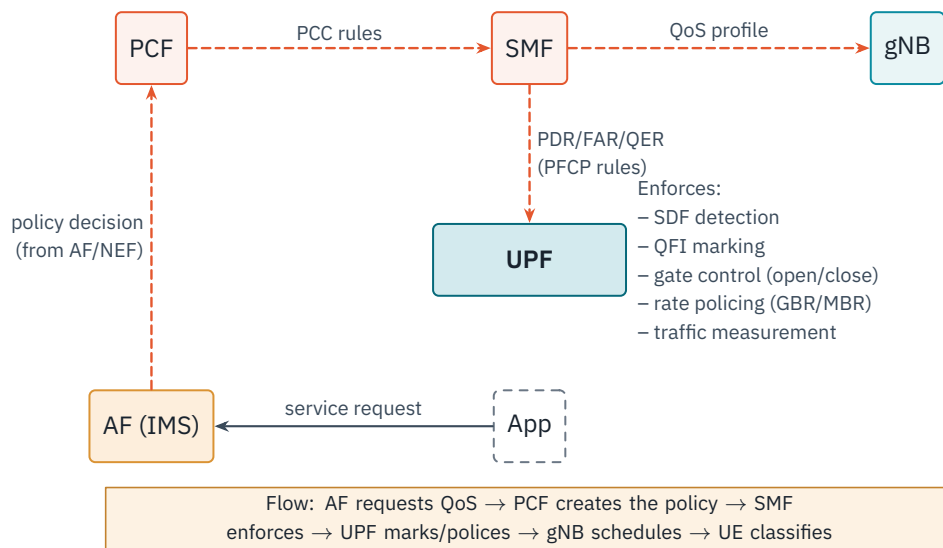


Figure 22.6 QoS policy and enforcement architecture

6 Bearer vs Flow Comparison

Side-by-Side Architecture

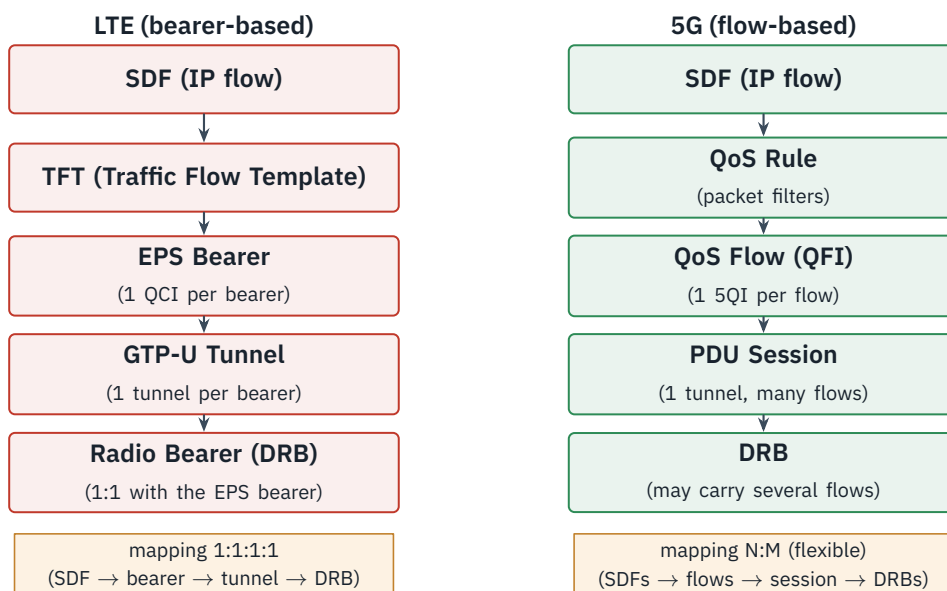


Figure 22.7 LTE bearer-based versus 5G flow-based QoS

Detailed Comparison Table

Aspect	LTE (Bearer-Based)	5G (Flow-Based)
QoS Unit	EPS Bearer	QoS Flow
Identifier	EBI (EPS Bearer ID)	QFI (QoS Flow ID, 6-bit)
QoS Class	QCI (1-9 + MCX)	5QI (1-90+)
Granularity	Coarse (1 QCI/bearer)	Fine (per-flow)
Tunnel	1 GTP-U per bearer	1 GTP-U per PDU session
Setup	NAS signaling per bearer	QoS rule install (lighter)
UL Classification	TFT (uplink packet filter)	QoS Rules
DL Classification	TFT at P-GW	SDF template at UPF
New QoS Class	New bearer (heavy signaling)	New flow (lightweight)
Sharing	No sharing between bearers	Flows share tunnel
Reflective QoS	Not supported	Supported (RQI)
Resource Types	GBR, Non-GBR	GBR, Non-GBR, DC-GBR

7 GBR vs Non-GBR vs Delay-Critical GBR

Three Resource Types in 5G

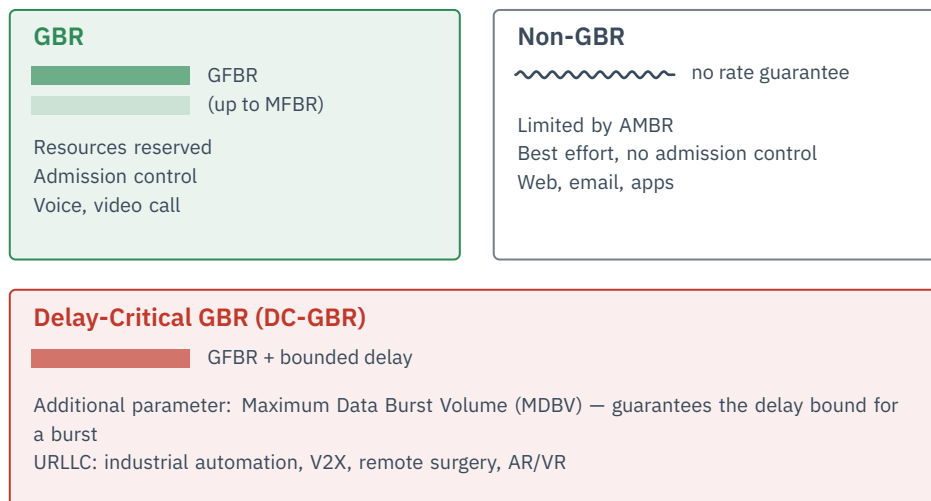


Figure 22.8 The three 5G QoS resource types

Parameters by Resource Type

Parameter	GBR	Non-GBR	DC-GBR
5QI	✓	✓	✓
ARP	✓	✓	✓
GFBP (Guaranteed Flow Bit Rate)	✓	✗	✓
MFBR (Maximum Flow Bit Rate)	✓	✗	✓
Session-AMBR	✗	✓	✗
Maximum Data Burst Volume	✗	✗	✓
Packet Delay Budget	✓	✓	✓ (strict)
Admission Control	Required	Not needed	Required
Resource Reservation	Yes	No	Yes

Rate Parameters Explained

- **GFBP = Guaranteed Flow Bit Rate**
 - the network **MUST** provide at least this rate
 - under congestion, GBR flows are protected up to GFBP
- **MFBR = Maximum Flow Bit Rate**
 - the network **MAY** provide up to this rate
 - traffic above MFBR is dropped or shaped

Example	GFBP	MFBR
VoLTE	40 kbps (AMR-NB codec rate)	80 kbps (with overhead margin)
Video call	1 Mbps (minimum acceptable quality)	4 Mbps (HD quality when resources allow)

8 AMBR (Aggregate Maximum Bit Rate)

■ AMBR Hierarchy

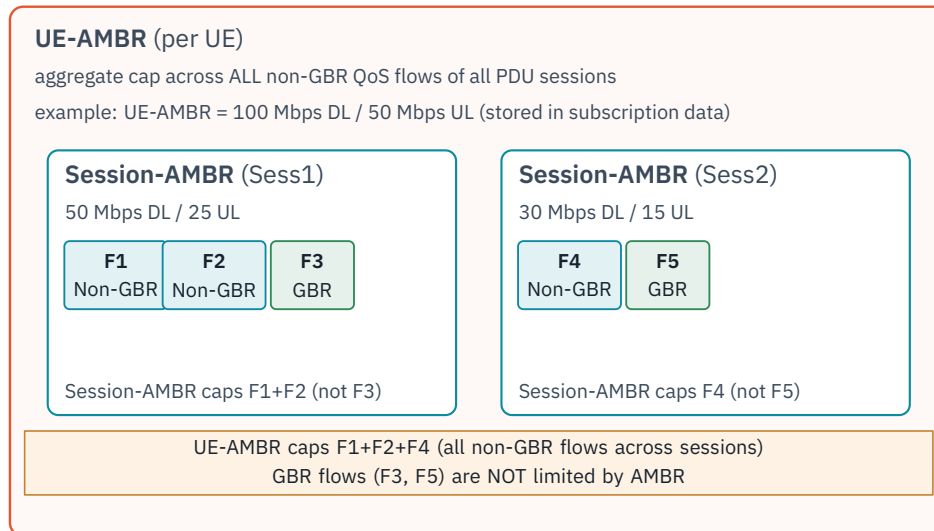


Figure 22.9 AMBR hierarchy: UE-AMBR above Session-AMBR; GBR flows are exempt

■ AMBR Enforcement Points

AMBR Type	Scope	Enforced At	Applies To
UE-AMBR	All sessions	gNB (RAN)	All Non-GBR flows, all sessions
Session-AMBR	Per PDU session	UPF	Non-GBR flows in that session

Key Rules:

- AMBR applies ONLY to Non-GBR flows
- GBR flows are limited by their own MFBR, not AMBR
- $UE-AMBR \geq \text{sum of Session-AMBRs (logically)}$
- Enforced separately for UL and DL

9 QoS Enforcement Architecture

Complete End-to-End QoS Signaling

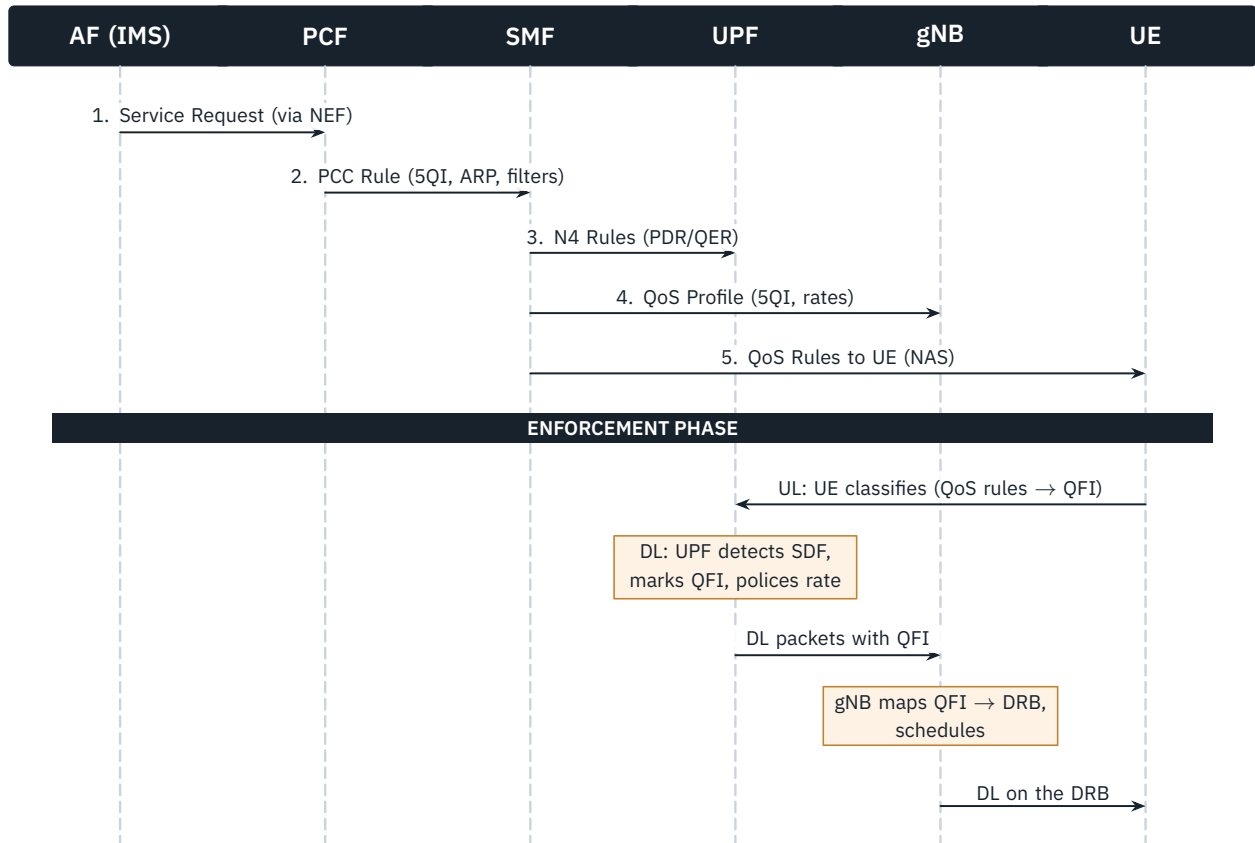


Figure 22.10 Complete end-to-end QoS signalling and enforcement

Summary of Enforcement Actions

Node	Uplink Actions	Downlink Actions
UE	Classify packets using QoS rules, mark QFI	Receive and process
gNB	Map QoS flows→DRBs, enforce UE-AMBR	Schedule based on 5QI priority, map DRB→flows
UPF	Verify QFI, enforce Session-AMBR	Detect SDF, mark QFI, gate/police/shape
SMF	Configure UPF (N4), signal gNB (N2), signal UE (N1)	Same
PCF	Generate PCC rules from policy	Same

Summary: Key Takeaways

Concept	Key Point
QoS Purpose	Different services need different network treatment
QCI (LTE)	9 standard classes, one per bearer
5QI (5G)	90+ classes, per QoS flow
Bearer (LTE)	1:1 mapping with QCI, separate GTP tunnel
QoS Flow (5G)	Many flows share one PDU session, lighter signaling
GBR	Guaranteed rate, admission control, resource reserved
Non-GBR	Best effort, limited by AMBR
DC-GBR	GBR + bounded delay (MDBV), for URLLC
AMBR	Aggregate cap on Non-GBR (UE-level and session-level)
Reflective QoS	UE derives UL QoS from DL (reduces signaling)
PCF	Policy brain, creates PCC rules
UPF	DL classifier and enforcer
gNB	Radio scheduler, DRB mapping, UE-AMBR

QCI vs 5QI Quick Reference

Feature	QCI (LTE)	5QI (5G)
Values	1-9 standard + 65-82 MCX	1-90+ standardized
Resource Types	GBR, Non-GBR	GBR, Non-GBR, DC-GBR
Priority Scale	1-9 (1=highest)	1-127 (1=highest)
Associated With	EPS Bearer	QoS Flow
Delay Range	50-300 ms	5-300 ms
New Category	—	Delay-Critical GBR
URLLC Support	Limited	Native (5ms delay, 10^{-5} loss)

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